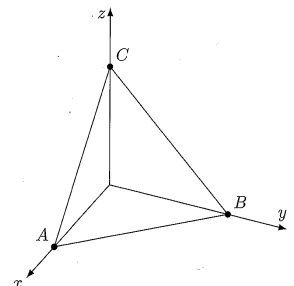
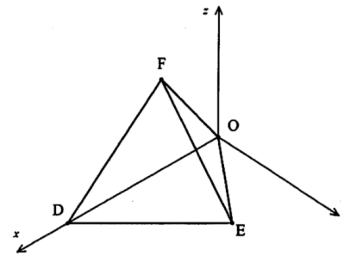
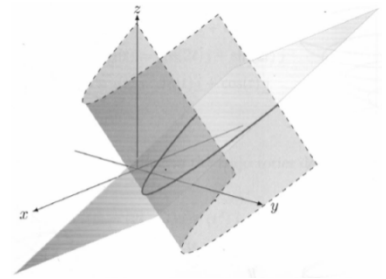


Vector WS 4

- The points A and B have position vectors relative to the origin O given by $\overrightarrow{OA} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$, $\overrightarrow{OB} = \begin{pmatrix} -1 \\ 3 \\ 4 \end{pmatrix}$.
 - Find an expression for the vector $\overrightarrow{AB}$ in the form of $x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$.
 - Show that the cosine of the angle between the vectors $\overrightarrow{OA}$ and $\overrightarrow{AB}$ is $\frac{4}{9}$.
 - Find the exact area of $\triangle OAB$.
- The base of a pyramid is the parallelogram ABCD with the vertices at points $A(2, -1, 3)$, $B(4, -2, 1)$, $C(a, b, c)$, $D(4, 3, -1)$. The apex (top) of the pyramid is located at $P(4, -4, 9)$.
 - Find the values of a , b and c .
 - Find the cosine of the angle between the vectors $\overrightarrow{AB}$ and $\overrightarrow{AD}$.
 - Find the exact area of the base of the pyramid.
 - Show that $\vec{v} = 6\hat{i} + 2\hat{j} + 5\hat{k}$ is perpendicular to both $\overrightarrow{AB}$ and $\overrightarrow{AD}$. Hence find a unit vector $\hat{v}$ that is perpendicular to the base of the pyramid.
 - Find the volume of the pyramid.
- Let the line l have the parametric equations $x = 3 - \lambda$, $y = 2 + 2\lambda$ and $z = 5 - 2\lambda$. Find the distance from the line to the point $\hat{R} = 3\hat{i} + 2\hat{j} - \hat{k}$.
- The faces of a tetrahedron OCEF are equilateral triangles of side length 1 unit. Its base ODE lies flat on the xy plane with two vertices at O and $D(1, 0, 0)$ with F above the xy plane.
 - Show that the coordinates of E are $(\frac{1}{2}, \frac{\sqrt{3}}{2}, 0)$.
 - Using vectors, prove the coordinates of the vertex F are $(\frac{1}{2}, \frac{\sqrt{3}}{6}, \frac{\sqrt{6}}{3})$.
- Show that an equation of the line that goes through the points $(8, -19, 13)$ and $(7, -15, 10)$ is $\vec{r} = (8 - \lambda)\hat{i} - (19 - 4\lambda)\hat{j} + (13 - 3\lambda)\hat{k}$.
 - Take an interval on the line $\hat{r}$ such that $-3 \leq \lambda \leq 7$ and find a point that divides that interval internally into a ratio of 3:2
- $\triangle ABC$ has vertices $A(1, -1, 0)$, $B(2, 1, -1)$ and $C(-1, 1, 2)$.
 - Use the scalar product to determine which of the angles of $\triangle ABC$ is a right angle.
 - Hence determine the area of $\triangle ABC$.
- If $\vec{a} = \hat{i} + 2\hat{j} - 3\hat{k}$, $\vec{b} = 5\hat{i} + 2\hat{j} - 4\hat{k}$, $\vec{c} = 2\hat{i} - \hat{j} - 4\hat{k}$, find the values of p and q such that $\vec{a} + p\vec{b} + q\vec{c}$ is parallel to the x -axis.
- The vector equation is given by $\vec{r} = (a \cos t)\hat{i} + (a \sin t)\hat{j}$, $a > 0$. Find the Cartesian equation of the curve and show that it is a circle.
- The points A, B and C have position vectors relative to a fixed point, O, such that $\overrightarrow{OA} = 2\hat{i} + \hat{j} + 2\hat{k}$, $\overrightarrow{OB} = 3\hat{i} - \hat{j}$ and $\overrightarrow{OC} = \hat{i} - 2\hat{j} - 2\hat{k}$. Prove that AC bisects OB at right angles.
- Graph the curve represented by each vector function.
 - $x = \sec t$, $y = \tan t$, $z = 3$
 - $x = 2 \sin t$, $y = 3 \cos t$, $z = 1$
- Consider the tetrahedron defined by $A(t, 0, 0)$, $B(0, t, 0)$ and $C(0, 0, t)$. Show that the angle between the plane ABC and the base is $\theta = \cos^{-1}(\frac{1}{\sqrt{3}})$.



12. Let P_1, P_2, P_3 and P_4 be the vertices of a regular tetrahedron circumscribed by the unit sphere. Let the position vectors of the vertices be $\vec{p}_1, \vec{p}_2, \vec{p}_3$ and $\vec{p}_4$ respectively.
- Show that $\vec{p}_1 + \vec{p}_2 + \vec{p}_3 + \vec{p}_4 = 0$
 - Prove that $\vec{p}_i \cdot \vec{p}_j = -\frac{1}{3}$ for all $i \neq j$.
 - Let P be any arbitrary point on the sphere. Show that the expression $|\overrightarrow{PP_1}|^2 + |\overrightarrow{PP_2}|^2 + |\overrightarrow{PP_3}|^2 + |\overrightarrow{PP_4}|^2$ is independent of the position of P .
13. a. Find the Cartesian equation of the curve defined parametrically by $\vec{r}(t) = (\cos t + \sin t)\vec{i} + (\cos t - \sin t)\vec{j}$
- b. Find the Cartesian equation of the curve defined parametrically by $\vec{r}(t) = \sqrt{2} \cos(t)\vec{i} + \sqrt{2} \sin(t)\vec{j}$
- c. Describe the difference in their position vectors as t varies.
14. The diagram below shows a parabolic cylinder $z = x^2 - y$ and a plane $z = y - 1$.
- Solve the surfaces simultaneously to show that $y = \frac{1}{2}(x^2 + 1)$. What is the geometric significance of this result?
 - Set $x = t$ for some $t \in \mathbb{R}$ and hence show that the intersection of the two surfaces has parametric representation $\vec{r}(t) = (t)\vec{i} + \frac{1}{2}(t^2 + 1)\vec{j} + \frac{1}{2}(t^2 - 1)\vec{k}$.
 - For the $\vec{k}$ -component above, the surface $z = y - 1$ was used. Is it incorrect to instead use the other surface $z = x^2 - y$?
 - Suppose that the condition $z \leq 4$ were introduced. Find a corresponding restriction for t and hence find the endpoints of the curve of intersection.



15. Consider the two mutually tangential spheres

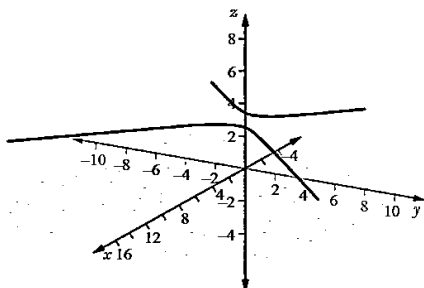
$$S_1: |\vec{v} - \vec{c}_1| = r_1$$

$$S_2: |\vec{v} - \vec{c}_2| = r_2$$

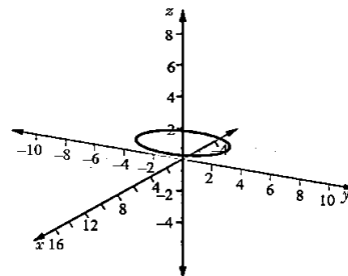
Show that the point of contact has position vector $\vec{p} = \frac{r_2 \vec{c}_1 + r_1 \vec{c}_2}{r_1 + r_2}$

Answer

- a. $\overrightarrow{AB} = -2\vec{i} + \vec{j} + 2\vec{k}$; c. $\frac{\sqrt{65}}{2}u^2$
- a. 6, 2, -3; b. $\frac{4}{9}$; c. $2\sqrt{65}u^2$; e. $24u^3$
- a. $\angle BAC = 90^\circ$; b. $3\sqrt{2}$
- $p = \frac{1}{6}, q = -\frac{11}{12}$
- $x^2 + y^2 = a^2$
- a.



b.



- a. $x^2 + y^2 = 2$; b. $x^2 + y^2 = 2$; c. The first one starts at (1, 1) and goes clockwise as t increases. The second one starts at $(\sqrt{2}, 0)$ and goes anti-clockwise as t increases.
- a. the parabola $y = \frac{1}{2}(x^2 + 1)$ is the projection of the curve of intersection onto the xy -plane; b. proving; c. no, it is not wrong to use the other surface. Choose $z = y - 1$ because the algebra will be simpler; d. $-3 \leq t \leq 3$. Intersections at $(-3, 5, 4)$ and $(3, 5, 4)$.